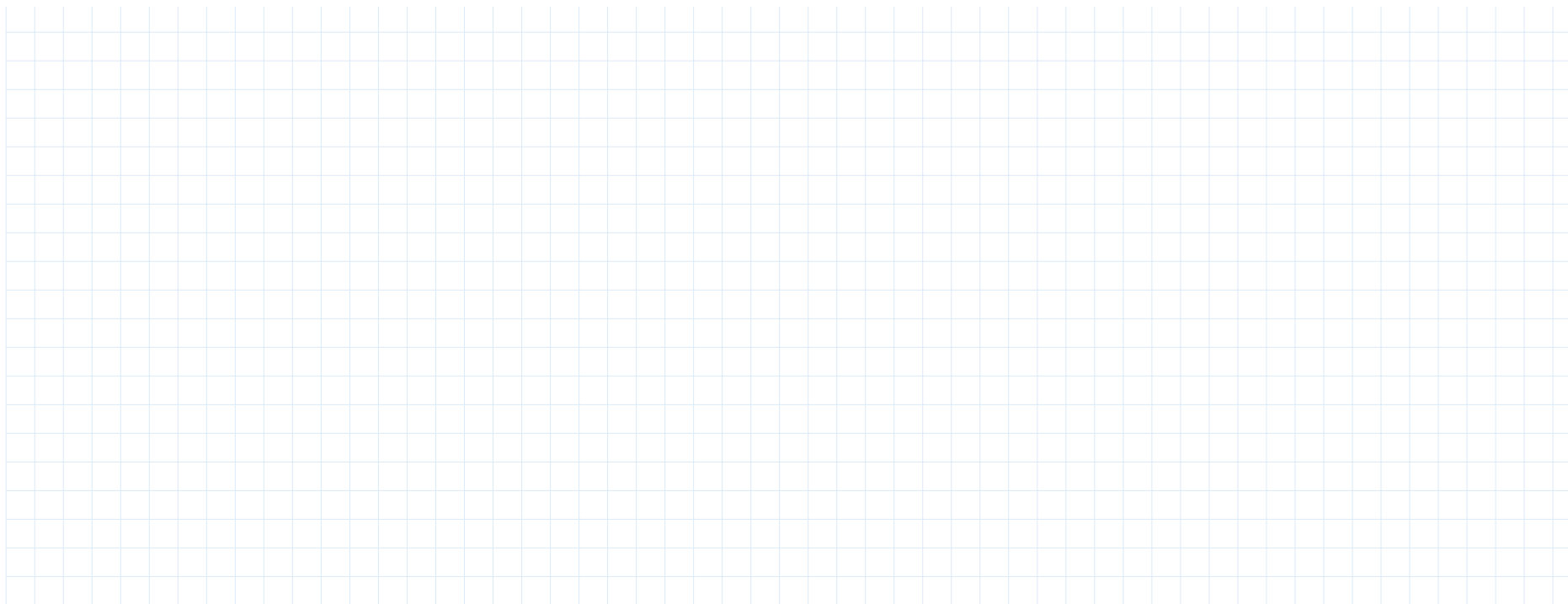


Division and a Slant Asymptote

Division writes $\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$, with $\deg R < \deg Q$. If $S(x)$ is nonconstant linear, $y = S(x)$ is the **slant asymptote**.

$$f(x) = \frac{x^2 + 2x + 4}{x - 1}.$$

Use long division. State the quotient, remainder, domain, and slant asymptote.



Use the Quotient to Describe the Ends

Divide and describe the end behavior of each function. Include zero coefficients for missing powers.

$$(a) \quad g(x) = \frac{2x^3 + x^2 + 3}{x^2 + 1} \qquad (b) \quad h(x) = \frac{x^3 + 2}{x - 1}.$$

Which has a slant asymptote? What does the other graph approach?



A Complete Rational Graph: Analyze

For

$$g(x) = \frac{x^2 - 1}{x^2 + x - 2},$$

find the original domain, simplified formula, hole, intercepts, and asymptotes. Find both one-sided limits at the vertical asymptote. We will draw the graph on the next page.



A Complete Rational Graph: Sketch

Sketch $g(x) = \frac{x^2 - 1}{x^2 + x - 2}$ using the previous page. Label the intercepts and hole. Use dashed asymptotes and an open circle at the hole.



For

$$h(x) = \frac{x^2 - 4}{x - 1},$$

find the domain, intercepts, and all asymptotes. Use division and one-sided limits to describe the branches. Does either branch cross the slant asymptote?



A Slant Graph: Sketch

Sketch $h(x) = \frac{x^2 - 4}{x - 1}$ using the previous page. Draw the vertical and slant asymptotes, label every intercept, and show both branches.

